Water Quality Metrics — Source of Truth

Scope: Reference for the six core parameters measured by the DataPod — Temperature, ORP, Dissolved Oxygen (DO), Electrical Conductivity (EC), pH, and Turbidity — across both freshwater and saltwater systems, and how they move together to fingerprint pollution events.

Core principle: No single parameter diagnoses a pollution event. Diagnosis comes from the *pattern of correlated change* across multiple parameters, and from separating that pattern from the water body's normal daily, tidal, and seasonal rhythms.

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1. Parameter Reference

Temperature

- **Measures:** Thermal energy of the water. Units: °C.
- **What drives it:** Solar input, air temperature, depth/stratification, tidal and current mixing, freshwater inflow, discharges.
- **Fresh vs. salt:** Freshwater bodies are smaller and respond faster to air temperature and runoff. Coastal/marine water is more thermally buffered but driven by tidal exchange and upwelling.
- **Pollution relevance:** Temperature is the *master variable* — it sets DO capacity, accelerates biological and chemical rates, and shifts EC readings. Thermal pollution (power-plant cooling water, hot-pavement stormwater) is itself an event type. Always interpret the other five parameters relative to temperature.

Dissolved Oxygen (DO)

- **Measures:** Oxygen dissolved in the water, available to aquatic life. Units: mg/L (concentration) and % saturation.
- **What drives it:** Temperature and salinity (set the saturation ceiling), photosynthesis (adds O_2 by day), respiration/decomposition (removes O_2), atmospheric exchange, turbulence.
- **Fresh vs. salt:** Saltwater holds roughly **20% less DO** than freshwater at the same temperature — marine systems sit closer to the hypoxia threshold by default.
- **Pollution relevance:** The single most important indicator of aquatic health. Organic pollution consumes oxygen (biochemical oxygen demand). Watch both the absolute value and the *shape of the daily curve* — large swings signal eutrophication; a sustained sag signals an oxygen-demanding discharge.

- **Thresholds (general):** >6 mg/L healthy · 4–6 mg/L stress · 2–4 mg/L hypoxic stress · <2 mg/L hypoxia · ~0 anoxia.

ORP (Oxidation-Reduction Potential)

- **Measures:** The water's net tendency to oxidize or reduce — effectively, the "electron pressure." Units: mV.
- **What drives it:** The dominant electron acceptor present. Oxygen-rich water reads strongly positive; as oxygen is consumed, the system shifts to nitrate, then manganese/iron, then sulfate, driving ORP downward and eventually negative.
- **Fresh vs. salt:** Healthy aerobic water typically reads **+200 to +400 mV**. Negative values indicate reducing, anoxic, or septic conditions. Seawater's high ionic strength makes absolute readings less comparable between sites — trend matters more than the raw number.
- **Pollution relevance:** ORP is the *early-warning twin of DO*. Because it responds to the chemical environment, it often begins falling before DO bottoms out. A drop toward or below zero is a strong signal of organic loading, sewage, or decomposition.

Electrical Conductivity (EC)

- **Measures:** The water's ability to carry current — a proxy for total dissolved ions/salts. Units: µS/cm (report temperature-corrected "specific conductance" at 25°C).
- **What drives it:** Dissolved ion content. Salinity dominates in marine water; in freshwater, EC reflects geology, runoff, road salt, and discharges.
- **Fresh vs. salt:** Freshwater and seawater differ by roughly **1,000×** in EC. This makes EC the primary tool for identifying the *source* of a change — freshwater input vs. saline input vs. an ionic discharge.
- **Pollution relevance:** Sudden EC shifts flag saltwater intrusion, road-salt runoff, industrial/chemical discharge, or sewage. In estuaries, EC normally tracks the tide — separate that rhythm from genuine events.

pH

- **Measures:** Acidity/alkalinity on a logarithmic scale of hydrogen-ion activity. Units: pH (unitless).
- **What drives it:** The CO_2–carbonate balance. Photosynthesis removes CO_2 and raises pH; respiration/decomposition adds CO_2 and lowers pH. Also affected by acidic/alkaline inputs.
- **Fresh vs. salt:** Freshwater (typically 6.5–8.5) is poorly buffered and naturally variable. Seawater (~8.0–8.2) is **strongly carbonate-buffered** — so any meaningful pH excursion in marine water is significant and worth investigating.
- **Pollution relevance:** Acidic inputs (acid mine drainage, certain industrial effluent) crash pH; alkaline inputs (concrete/construction washwater, some industrial discharge) raise it. Large

daily pH swings indicate biological productivity, not a discharge.

Turbidity

- **Measures:** Water clarity — light scattered by suspended particles (sediment, plankton, organic matter). Units: NTU or FNU.
- **What drives it:** Sediment runoff, shoreline erosion, bottom resuspension by currents/waves/dredging, and algal cells.
- **Fresh vs. salt:** Rivers and estuaries are naturally more turbid and event-driven (storms, dredging). Open coastal water is clearer; resuspension peaks at maximum tidal current.
- **Pollution relevance:** A turbidity spike flags storm runoff, construction/erosion, or dredging. Critically, **particles carry adsorbed pollutants** — phosphorus, metals, hydrocarbons, pathogens — so a turbidity spike is also a proxy for contaminant loading. High turbidity also suppresses photosynthesis by blocking light.

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2. Baseline Reference Ranges

Parameter	Healthy Freshwater	Brackish / Estuarine	Healthy Seawater
Temperature	Climate/season-dependent	Climate/season-dependent	Climate/season-dependent
Dissolved Oxygen	6–11 mg/L (>80% sat.)	5–9 mg/L	5–8 mg/L (~7–8 at 100% sat.)
ORP	+200 to +400 mV	+150 to +350 mV	+200 to +400 mV
Conductivity (EC)	50–1,500 µS/cm	~1,000–35,000 µS/cm	~45,000–55,000 µS/cm (~35 PSU)
pH	6.5–8.5	7.0–8.3	7.8–8.3
Turbidity	<5–25 NTU	5–100+ NTU	<5–10 NTU

Ranges are general guidance. Establish a site-specific baseline before treating deviations as events — local geology, season, and tidal stage shift “normal” substantially.

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3. How the Parameters Interrelate

Understanding these couplings is what turns six independent readings into a diagnostic system.

- **Temperature → DO (inverse, two ways):** Warmer water physically holds less oxygen *and* speeds the respiration/decomposition that consumes it. A temperature rise should always be paired with an expected DO decline.

- **Temperature → EC:** Ion mobility rises with temperature; raw EC increases ~2% per °C. Always report temperature-corrected specific conductance so EC changes reflect chemistry, not warming.
- **DO ↔ ORP (tightly coupled):** Oxygen is the dominant electron acceptor — abundant DO drives ORP strongly positive. As DO is consumed, ORP declines; once oxygen is exhausted, ORP goes negative. ORP often moves *before* DO hits bottom, giving lead time.
- **pH ↔ DO (the photosynthesis/respiration couple):** Photosynthesis removes CO_2 (raises pH) and releases O_2; respiration adds CO_2 (lowers pH) and consumes O_2. pH and DO therefore rise and fall *together* on the daily cycle. **In-phase daily swings of pH and DO are the fingerprint of biological productivity.**
- **Turbidity → DO/pH:** Suspended particles block light, suppressing photosynthesis and dampening daytime DO/pH gains. Organic-rich turbidity also adds direct oxygen demand.
- **Turbidity as a contaminant carrier:** Sediment transports adsorbed nutrients, metals, and pathogens — a turbidity spike can stand in for a loading event even when chemistry looks stable.
- **EC as a source fingerprint:** Because fresh and salt water differ ~1,000× in EC, the *direction* of an EC change tells you whether freshwater (rain, runoff, groundwater) or saline/ionic input (tide, intrusion, discharge) is driving the event.

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4. Pollution Event Signature Matrix

Arrows show the expected direction of change relative to baseline. ↑↑ / ↓↓ = strong, ↑ / ↓ = moderate, ↕️ = large swing or variable, → = little change.

Event	Temp	DO	ORP	EC	pH	Turbidity	Primary Signature
Sewage / sanitary discharge	→/↑	↓↓	↓↓	↑	↓	↑	DO crash **+** ORP crash **+** EC rise **+** turbidity rise, *not* tied to time of day
Algal bloom /		↕️ (super-sat. midday,			↕️ (high		Large *in-phase* daily DO + pH oscillations;

eutrophication	↑	crash pre-dawn)	↕️	→	midday)	↑	pre-dawn DO minimum is the danger window
Stormwater / urban runoff	↕️	↓	↓	↓ (marine) / ↕️ (fresh)	↕️	↑↑	Sharp turbidity spike coincident with rainfall; EC shifts (drops in marine, may spike in fresh from road salt)
Industrial / chemical discharge	↕️	↕️	↕️↕️	↑↑	↕️↕️	↕️	Abrupt *step-changes* in EC and/or pH and/or ORP with no diel or tidal explanation
Thermal discharge	↑↑	↓	→	→	→	→	Temperature rise with proportional DO decline; other chemistry flat
Saltwater intrusion	→	→	→	↑↑	→/↑	→	EC rise correlated with tidal phase, drought, or sea-level conditions
Acidic input (acid mine drainage / spill)	→	↕️	↑	↑	↓↓	↑	pH crash paired with EC rise (dissolved metals/sulfate)
Hypoxia / fish-kill conditions	often ↑	→0	↓↓ (negative)	→	↓	↕️	DO and ORP both bottoming out together

Reading the matrix: The diagnostic value is in the *combination*. DO falling alone could be warm weather. DO **and** ORP falling together while EC and turbidity rise is a far more specific signal pointing to an organic/sewage source.

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5. Separating Natural Cycles from Pollution Events

Rule out the water body's normal rhythms before declaring an event.

- **Diel (24-hour) cycle:** DO, pH, and ORP swing predictably each day — peaking mid-afternoon, bottoming pre-dawn — driven by photosynthesis and respiration. Temperature follows the sun. A *smooth, repeating* daily oscillation is biology, not pollution.
- **Tidal cycle (estuarine/coastal):** EC, temperature, and turbidity oscillate with the tide. Turbidity often peaks at maximum current (resuspension). An EC change that repeats on the tidal clock is mixing, not a discharge.
- **Seasonal cycle:** Temperature, stratification, runoff, and bloom potential shift across the year. Compare any reading to the same season's baseline.
- **The test for a real event:** A pollution event typically appears as a **step-change or sustained excursion that breaks the expected diel/tidal rhythm** — not a smooth oscillation. Confirm it shows up as a *correlated* shift across multiple parameters consistent with one of the signatures above.

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6. Sensor & Data-Quality Caveats

A trend is only as trustworthy as the sensor producing it. Rule out instrument drift before calling an event.

- **Calibration drift:** pH, DO, and ORP probes drift over time. A slow one-directional trend across weeks may be drift, not water chemistry — cross-check against calibration history and class-balanced expectations.
- **ORP reference offset:** ORP values depend on the electrode's reference type (Ag/AgCl vs. standard hydrogen electrode). Record which reference is used so values are comparable across DataPods and over time. Treat ORP *trend* as more reliable than absolute value.
- **EC temperature compensation:** Always report specific conductance (25°C-corrected). Uncorrected EC will track temperature and produce false signals.
- **Turbidity units:** NTU (white-light) and FNU (infrared) are *not interchangeable*. Standardize on one across the fleet.
- **DO sensor fouling:** Optical DO sensors drift with biofilm growth. Pair DO trends with the antifouling/cleaning schedule.

- **Biofouling, broadly:** All submerged sensors foul. A gradual multi-parameter drift after a long deployment is a fouling/maintenance flag first, an environmental signal second.

Maintain this as a living document. As DataPod deployments build site-specific baselines, append local “normal” ranges and any site-specific event signatures observed in the field.